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-- Database is Not Case-Sensitive
-- For creating a database, use the following command:
create database student_db;
create database teacher_db;
create database employee_db;

-- Delete a database:
-- drop database student_db;
drop database employee_db;

-- Use the database
use student_db;

/*
1. For creating a table:
create table table_name (
    column1_name column1_datatype constraint,
    column2_name column2_datatype constraint,
    column3_name column3_datatype constraint
);

#Constraint is optional but recommended to use.
-- Constraints are used to define rules for the data in a table.
-- Constraints are used to limit the type of data that can go into a table.

1. NOT NULL: Means that the column cannot have a NULL value.
2. UNIQUE: Means that the column must have a unique value.
3. PRIMARY KEY: A combination of a NOT NULL and UNIQUE. Uniquely identifies
each row in a table.
4. FOREIGN KEY: Uniquely identifies a row/record in another table.
5. CHECK: Ensures that all values in a column satisfy a specific condition.
6. DEFAULT : Sets a default value for a column when no value is specified.

#PRIMARY KEY:
This is the most important constraint. It is used to uniquely identify each row
in a table.
A primary key must contain unique values, and it cannot contain NULL values.
Table won't be created without PRIMARY KEY

#DataTypes:
- INT: A normal-sized integer that can be signed or unsigned.
- Float: A floating point number that can be signed or unsigned.
- Double: A double-sized floating point number that can be signed or unsigned.
- char(n): A fixed-length string that can contain letters, numbers, and special
characters.

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- varchar(n): A variable-length string that can contain letters, numbers, and special characters.
- date: A date value in the format YYYY-MM-DD.
- time: A time value in the format HH:MM:SS.
- datetime: A date and time value in the format YYYY-MM-DD HH:MM:SS.

#Note: We mostly use INT, VARCHAR, DATE, and DATETIME data types in our tables.
*/

```
CREATE TABLE student (  
    id INT PRIMARY KEY,  
    name VARCHAR(50) NOT NULL,  
    section CHAR(2) NOT NULL,  
    age INT CHECK (age > 18),  
    admission_date DATE,           -- Stores only the date (e.g., 2025-05-26)  
    class_start_time TIME,        -- Stores only the time (e.g., 08:30:00)  
    registration_datetime DATETIME -- Stores both date and time (e.g., 2025-05-26 08:30:00)  
);
```

-- 2.Insert data into the table:

-- insert into table_name (id, name, section, age)

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insert into student (id, name, section, age) values  
(1, "John Doe", 'A1', 20),  
(2, "Jane Smith", 'B2', 22),  
(3, "Alice Johnson", 'C3', 19),  
(4, "Bob Brown", 'D4', 21);
```

-- 3.Show/Print:

-- select * from table_name;

-- select column1, column2 from table_name;

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select * from student;
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select name, age from student;
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-- select * from student where age > 20;

-- 4.Update data in the table:

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update student (Table's Name)
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set name = "Bear Grylls" where id = 1;
```

-- we will update with PRIMARY KEY mostly

-- 5. Delete data from the table:

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delete from student (Table's Name)
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```
where id = 2;
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--6.Delete the table:
drop Table student;

--7.Alter table:
-- 1. Add a new column to the table:
alter table student
add column email varchar(50) not null;
-- 2. Drop a column from the table:
alter table student_
drop column email;
-- 3. Rename a column in the table:
alter table student
rename column name to full_name;
-- 4. Change the data type of a column in the table:
alter table student
modify column age float;
-- 5. Rename the table:
alter table student
rename to student_info;
-- 6. Add a primary key to the table:
alter table student
add primary key (id);

```

Example:

```

-- Step 1: Create the database
CREATE DATABASE CompanyDB;
USE CompanyDB;

-- Step 2: Create the EMPLOYEE table
CREATE TABLE EMPLOYEE (
    EMP_ID INT PRIMARY KEY,
    FIRST_NAME VARCHAR(50),
    LAST_NAME VARCHAR(50),
    DEPARTMENT VARCHAR(50),
    SALARY INT,
    HIREDATE DATE
);

```

```
-- Step 3: Insert values into EMPLOYEE table
INSERT INTO EMPLOYEE (EMP_ID, FIRST_NAME, LAST_NAME, DEPARTMENT, SALARY,
HIREDATE) VALUES
(1, 'JOHN', 'DEO', 'HR', 50000, '2022-01-10'),
(2, 'JANE', 'SMITH', 'IT', 60000, '2023-03-12'),
(3, 'ALICE', 'JOHNSON', 'HR', 35000, '2020-05-03'),
(4, 'BOB', 'BROWN', 'FINANCE', 62000, '2021-05-15'),
(5, 'CHARLIE', 'DAVINS', 'IT', 75000, '2017-04-07');

-- Step 4: Add a column 'GENDER' with default value 'Male'
ALTER TABLE EMPLOYEE ADD GENDER VARCHAR(10) DEFAULT 'Male';

-- Step 4.1: Set 'GENDER' value explicitly for existing rows (if needed)
UPDATE EMPLOYEE SET GENDER = 'Male';

-- Step 5: Update the salary of EMP_ID 5 to 65,000
UPDATE EMPLOYEE SET SALARY = 65000 WHERE EMP_ID = 5;

-- Step 6: Delete employees from FINANCE department
DELETE FROM EMPLOYEE WHERE DEPARTMENT = 'FINANCE';

-- Step 7: Rename column HIREDATE to JOINDATE
ALTER TABLE EMPLOYEE CHANGE HIREDATE JOINDATE DATE;
```
